University of Toronto
Scarborough Campus
April 16, 2024

CSC C63 Final Exam
Instructor: Vassos Hadzilacos

Aids allowed: One 8.5×11 handwritten, non-photocopied 'cheat sheet' (may be written on both sides)

Duration: Three hours

READ THE INSTRUCTIONS CAREFULLY

- There should be 17 pages in this exam booklet, including this cover page.
- Answer all questions.
- Put all answers in this booklet, in the spaces provided.
- For rough work, use the backs of the pages; ***these will not be graded***.
- The last three pages are blank and can be used for rough work or for overflow. ***Do not detach them*** from the booklet. If you use them for overflow, you must ***clearly indicate this*** on the page(s) containing the question(s) whose answer(s) you are providing in the overflow pages.
- In your answers you may use any result discussed in this course or its prerequisites merely by naming or describing it. In particular, you may use any of the facts regarding undecidable, unrecognizable, and NP-complete problems listed on page 2.
- Good luck!

LIST OF UNDECIDABLE OR UNRECOGNIZABLE SETS:

- $\textsc{Univ} = \{\langle M, x\rangle$: TM M accepts input $x\}$ is undecidable but recognizable; thus $\overline{\textsc{Univ}}$ is unrecognizable.
- $\textsc{Halt} = \{\langle M, x\rangle$: TM M halts on input $x\}$ is undecidable but recognizable; thus $\overline{\textsc{Halt}}$ is unrecognizable.
- $\text{HET} = \{\langle M\rangle$: TM M halts on empty tape$\}$ is undecidable but recognizable; thus $\overline{\text{HET}}$ is unrecognizable.
- $\overline{\textsc{Empty}} = \{\langle M\rangle$: TM M accepts some input$\}$ is undecidable but recognizable; thus $\textsc{Empty}$ is unrecognizable.
- $\textsc{Fin} = \{\langle M\rangle$: TM M recognizes a finite language$\}$ and $\overline{\textsc{Fin}}$ are both unrecognizable.
- $\textsc{Reg} = \{\langle M\rangle$: TM M recognizes a regular language$\}$ and $\overline{\textsc{Reg}}$ are both unrecognizable.

LIST OF NP-COMPLETE PROBLEMS:

- SAT/CNF-SAT/3SAT
  Instance: $\langle F\rangle$, where F is a propositional formula/CNF formula/CNF formula with three-literal clauses.
  Question: Is F satisfiable?

- CLIQUE/INDEPENDENT SET/VERTEX COVER
  Instance: $\langle G, k\rangle$, where G is an undirected graph and k is a positive integer.
  Question: Does G have a clique/independent set/vertex cover of size k?

- EXACT COVER
  Instance: $\langle U, \mathcal{C}\rangle$, where U is a set and $\mathcal{C}$ is a set of subsets of U.
  Question: Is there a subset of $\mathcal{C}$ that contains pairwise disjoint sets whose union is U?

- 3-DIMENSIONAL MATCHING
  Instance: $\langle A, B, C, M\rangle$, where A, B, C are disjoint sets such that $|A| = |B| = |C| = n$, and $M \subseteq A \times B \times C$.
  Question: Is there a subset of M that is a matching of size n?

- COLOURING
  Instance: $\langle G, k\rangle$, where $G = (V, E)$ is an undirected graph and k is a positive integer.
  Question: Is there a function $f : V \to [1..k]$ so that if $\{u, v\} \in E$ then $f(u) \neq f(v)$?

- DIRECTED/UNDIRECTED HAMILTONIAN CYCLE/PATH
  Instance: $\langle G\rangle$, where G is a directed/undirected graph.
  Question: Does G have a Hamiltonian cycle/path?

- SUBSET SUM
  Instance: $\langle A, k\rangle$, where A is a sequence of positive integers and k is a positive integer.
  Question: Is there a subsequence of A whose elements add up to k?

- PARTITION
  Instance: $\langle A\rangle$, where A is a sequence of positive integers.
  Question: Does A have a subsequence whose elements add up to half of the sum of the elements of A?

- ZERO-ONE LINEAR PROGRAMMING
  Instance: $\langle A, \mathbf{b}\rangle$, where A is an $m \times n$ matrix of integers, and $\mathbf{b} \in \mathbb{Z}^m$ is an m-dimensional vector of integers.
  Question: Is there a vector $\mathbf{x} \in \{0, 1\}^n$ such that $A\mathbf{x} \leq \mathbf{b}$?

QUESTION 1. (56 marks)

For each of the following statements indicate whether it is true, false, or open, and briefly justify your answer. Select "open" if the statement being true or it being false would resolve some open question discussed in this course, such as $\mathbf{P} \stackrel{?}{=} \mathbf{NP}$, $\mathbf{NP} \stackrel{?}{=} \mathbf{coNP}$, $\mathbf{P} \stackrel{?}{=} \mathbf{PSPACE}$, etc. In this case, justify your answer by stating ***which*** open question would be resolved by the statement and ***why***.

a. The language $\{\langle M\rangle \colon M$ is a TM with at most 100 states and at most 10 tape symbols, and $\mathcal{L}(M)$ is infinite$\}$ is undecidable.

Circle the correct answer: True | False | Open

Justification:

b. Every cofinite language is decidable. (Recall that a language is cofinite if its complement is finite.)

Circle the correct answer: True | False | Open

Justification:

c. If M is a Turing machine that loops on an infinite number of inputs then $\mathcal{L}(M)$ is undecidable.

Circle the correct answer: True | False | Open

Justification:

[continued on the next page]

d. $\textsc{Univ} \leq_m 3\textsc{Sat}$.

Circle the correct answer: True | False | Open

Justification:

e. The language $L = \begin{cases} \{\langle M \rangle\colon M \text{ has more states than tape symbols}\}, & \text{if } \mathbf{P} = \mathbf{NP} \\ \{\langle M \rangle\colon \mathcal{L}(M) \text{ contains no string of even length}\}, & \text{otherwise} \end{cases}$ is undecidable.

Circle the correct answer: True | False | Open

Justification:

f. $3\textsc{Sat} \leq_T^p 2\textsc{Sat}$. (Recall that $\leq_T^p$ is the Cook-reduces relation, i.e., the polytime-Turing-reduces relation.)

Circle the correct answer: True | False | Open

Justification:

[continued on the next page]

g. 3Sat $\in$ **coNP**.

Circle the correct answer: True | False | Open

Justification:

h. **NPSPACE** is closed under complementation. (Recall that **NPSPACE** is the set of decision problems that can be solved by nondeterministic Turing machines in $O(n^k)$ space, where k is a constant.)

Circle the correct answer: True | False | Open

Justification:

QUESTION 2. (20 marks)

Let A_1, A_2 be recognizable languages, such that $A_1 \cup A_2$ and $A_1 \cap A_2$ are both decidable. Prove that A_1 is decidable. (A_2 is also decidable by a symmetric argument, which you do not need to give.)

Hint: Make sure you use all of the assumptions about A_1 and A_2.

PROOF:

QUESTION 3. (10 marks)

Consider the language

$$B = \{\langle M \rangle \colon \mathcal{L}(M) \text{ and } \overline{\mathcal{L}(M)} \text{ are both infinite}\}.$$

($\overline{\mathcal{L}(M)} = \Sigma^* - \mathcal{L}(M)$, where Σ is the input alphabet of Turing machine M.) State the lowest Σ_i or Π_i set of the arithmetic hierarchy in which you can place B, and justify your answer by providing an appropriate formula that represents B.

ANSWER:

QUESTION 4. (30 marks)

Recall that a string is a ***palindrome*** if it reads backwards the same as forwards; for example, the empty string and 1001001 are palindromes; 1011 is not a palindrome. Let C be the following language

$$C = \big\{\langle M\rangle \colon \mathcal{L}(M) = \big\{x \colon x \text{ is a palindrome over the input alphabet of } M\big\}\big\}.$$

For each of C and its complement $\overline{C}$, state whether it is (a) decidable, (b) undecidable but recognizable, or (c) unrecognizable. Prove your answers.

ANSWER:

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QUESTION 5. (30 marks)

Clustering is an important problem that arises in statistics and machine learning. Given a set A of "points" (objects of interest) and the distances between them (measuring "similarity" between objects), we want to assign the objects into a small number of pairwise disjoint sets called "clusters", so that points in the same cluster are close to each other (i.e., they represent "similar" objects). We can formalize this as the decision problem CLUSTERING, defined as follows:

Instance: $\langle A, d, t, k\rangle$, where A is the set of points; d: $A \times A \to \mathbb{Z}^+$ is a function that specifies the distance $d(a, b)$ between the points $a, b \in A$, where we require that (i) $d(a, b) = 0$ if and only if $a = b$ and (ii) $d(a, b) = d(b, a)$; t is a positive integer that defines the maximum distance allowed between two points assigned to the same cluster; and k is a positive integer that defines the number of clusters.

Question: Can ***each*** element of A be assigned to ***exactly one*** of k sets $A_1, \ldots, A_k$ so that if a and b are assigned to the same set then $d(a, b) \leq t$?

Prove that CLUSTERING is **NP**-complete. For your reduction use one of the following problems: DIRECTED HAMILTONIAN CYCLE, COLOURING, INDEPENDENT SET, or ZERO-ONE LINEAR PROGRAMMING.

PROOF:

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QUESTION 6. (34 marks)

Read both parts of this question before answering either; part (b) can be done even if you don't do part (a).

a. (16 marks) Let $G = (V, E)$ be a directed graph, and $\mathbf{wt}\colon E \to \mathbb{Z}^+$ be a function that assigns a positive integer to each edge, the "weight" of that edge. A ***simple path*** p of G is a sequence of ***distinct*** nodes $p = u_1, u_2, \ldots, u_k$ such that $(u_i, u_{i+1}) \in E$ for each $i \in [1..k-1]$. The ***weight*** of such a path p, denoted $\mathbf{wt}(p)$, is the sum of the weights of the edges on it, i.e., $\mathbf{wt}(p) = \sum_{i=1}^{k-1} \mathbf{wt}(u_i, u_{i+1})$. Path Weight is the decision problem defined below:

Instance: $\langle G, \mathbf{wt}, \ell \rangle$, where $G = (V, E)$ is a directed graph, $\mathbf{wt}\colon E \to \mathbb{Z}^+$, and $\ell \in \mathbb{Z}^+$.
Question: Is there a simple path of G that has weight ***at least*** ℓ?

Prove that Path Weight is **NP**-complete.

Hint: Consider instances of Path Weight with restricted edge weights.

PROOF:

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b. (18 marks) An optimization problem related to Path Weight is to find a ***maximum***-weight simple path of a given graph. More precisely, the Max Weight Path problem is:

Instance: $\langle G, \mathbf{wt} \rangle$, where $G = (V, E)$ is a directed graph and $\mathbf{wt}$: $E \to \mathbb{Z}^+$.
Output: A simple path p of G such that for every simple path p' of G, $\mathbf{wt}(p) \geq \mathbf{wt}(p')$.

Give a Cook reduction (i.e., a polynomial-time Turing reduction) of the Max Weight Path problem to the Path Weight decision problem. In other words, describe a polynomial-time algorithm for Max Weight Path that uses a black box PW-Oracle that solves Path Weight, where each use of the black box is assumed to take one unit of time. First describe your reduction in clear English and then in high-level pseudocode that involves calls to PW-Oracle. Justify why your reduction is a polynomial-time one; do ***not*** justify that it is correct.

ANSWER:

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